

Contact

Bengaluru, India
9003037804 (Mobile)
pulkitindia5@gmail.com

www.linkedin.com/in/
buddywhitman (LinkedIn)
buddywhitman.tech (Portfolio)
github.com/buddywhitman (Other)
orcid.org/0000-0003-4078-1780
(Other)

Top Skills

Session Initiation Protocol (SIP)
SIP Trunking
Big Data

Languages

Tamil (Elementary)
Hindi (Native or Bilingual)
German (Elementary)
English (Native or Bilingual)
French (Elementary)
Kannada (Limited Working)

Certifications

AWS Cloud Solutions Architect
ISC2 Systems Security Certified
Practitioner (SSCP)

Publications

Technical Validation of a Multimodal
Emotion - Adaptive Biofeedback
System for Autonomic Regulation
Physics-Informed Stochastic
Receding Horizon Control for
Autonomous Energy Management in
Solar Racing

Patents

A System and Method for Adaptive
Interface in Electric Vehicle Control
and Monitoring Applications
A System and Method for Long-
Range Telemetry and Observability
for Solar Electric Vehicles

Pulkit Kumar

building multilingual agentic AI, solar racing EVs & native
computeruse for linux
Bengaluru, Karnataka, India

Summary

Shipped my first website and PCB at the age of 9. Today, I am a pre-final year undergrad building on the intersection of high-performance multi-agent systems and deterministic edge hardware. My work is divided into two core domains:- AI-native Distributed Systems: I architect agentic workflows and production-grade inference layers that serve 250K+ users. From implementing cross-modal recommender engines for wearables to deploying fault-tolerant ticketing systems handling 10x burst loads, the focus is on high-throughput reliability.- Safety-Critical Embedded Systems: As the Electronics Head for the university's solar racing team, I engineered the firmware for an Arm Cortex M7-based custom ECU. I've filed 3+ patents for adaptive EV interfaces and long-range telemetry systems, achieving a 55% reduction in thread latency for real-time control. Stack Highlights: AI/Software: LangGraph, PyTorch, Redis, Kubernetes, LiveKit, and Distributed Microservices. Embedded/ Hardware: MISRA C/C++, FreeRTOS, CAN Bus, FPGA (Verilog), and Edge AI (TensorRT). I am driven by an obsession for performance- whether I'm optimizing a digital twin for an endurance race or scaling cloud-native services to 99.99% uptime. If you are building high-stakes distributed/embedded systems, or anything on the edge- I'd love to connect!

Experience

Fettle
SDE (AI)
January 2026 - Present (6 months)
Hyderabad

Building voice AI agents for hospital customer experience workflows, and recommendation systems for corporate healthcare

Guaq AI
Co-Founder

A System and Method for Automated
Robotic Solar Panel Cleaning with
Obstacle Detection and Removal

November 2025 - Present (8 months)

Building intelligent CRMs and automation platforms for businesses not exploiting LLMs and data yet. Scaled to 6+ clients across hospitality, real estate, events & retail, with ~\$3K MRR in 2 months.

SolarMobil Manipal

Electronics Head

January 2024 - Present (2 years 6 months)

Manipal, India

- Implemented TouchGFX and FreeRTOS-based firmware for Cortex-M7 ECU and display units, improving thread latency by 210ms.
- Designed RF telemetry + weather/spatial fusion cloud dashboards for real-time race strategy, contributing to the best solar endurance performance in the country.
- Introduced version control and HIL pipelines to improve schedule adherence to 90%+

Teachafy Labs

Machine Learning Engineer

July 2025 - December 2025 (6 months)

Bengaluru

Developed Google ADK-based agentic orchestration and WhatsApp business chatbots to inject natural language workflows into ERP platforms, saving 5000+ hrs/week across 17+ clients.

Symbionic

Embedded Software Engineer

April 2025 - June 2025 (3 months)

Chennai, Tamil Nadu, India

Developed firmware around custom ESP32-based hardware for EMG/EEG signal acquisition & spatiotemporal fusion

Manipal Institute of Technology

Undergraduate Research Assistant

April 2025 - June 2025 (3 months)

Manipal, India

Interfaced biomedical probes (ECG, temp, GSR) and innovative respiration measurement using IMUs with digital signal processing & IoT-derived sensor fusion for hospital management & biofeedback digital twin studies. Reduced hospital staff response times to <5 min, enhancing patient survivability by 35%

+

Foodswipe

Co-Founder

January 2025 - June 2025 (6 months)

Bengaluru

Outsourcing food choices to content-based & collaborative recommender systems that actually understand what you'd like to eat. A/B testing rated positive customer experience at 85%+

DCPR AI

Software Engineer

July 2024 - December 2024 (6 months)

Bengaluru, Karnataka, India

Built document ingestion pipelines to automate PDF batch processing from GCS buckets using intelligent OCR inference at scale. Ingested 250TB+ poorly scanned docs through distributed Redis locking, concurrent Python workers, ELK monitoring, and Kubernetes HPA. Replaced manual processing, slashing TTM by 80%+ and saving \$600+/mo in data entry costs.

Developed RAG systems for legal and zoning documents with OpenAI Assistants API & Pinecone vector DB, achieving precise document & page citations with 92% accuracy, reducing legal compliance timelines for real estate developers in Mumbai by almost half.

Krop AI

Embedded Engineer

September 2024 - November 2024 (3 months)

Manipal, Udupi, Karnataka, India

Developed control & sensing systems + Jetson AI vision/perception pipeline for agritech systems.

CoreEL Technologies

Digital Design Engineer

April 2024 - June 2024 (3 months)

Bengaluru, Karnataka, India

Developed RTL design spec of high-speed CAN bus & I2C controllers, optimizing memory/macros for Zynq 7000 FPGA.

Collaborated with synthesis & verification teams to ensure QA standards were met. Integrated relevant IP blocks and gained expertise with the Vitis software suite.

Education

Manipal Institute of Technology

Bachelor of Technology - BTech, Electrical, Electronics and Communications Engineering · (July 2023 - July 2027)

St. Francis De Sales Public School

High School Diploma, Computer Science · (June 2017 - May 2023)

Primrose Schools Chennai

Middle School, Science · (2010 - 2017)